

On the Construction of Configurations n_3

Inaugural Dissertation

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Dedicated to his highly honored teacher

Professor Dr. Rosanes

by the author.

The construction of configurations n_3 has so far been treated in detail only in special cases. Martinetti recognized that, in general, it leads to a problem of the second degree, for example to the problem of determining the double elements of two incident point rows, and he also gave an indication of the path leading to this.¹ The subject is, however, interesting enough to deserve a more detailed treatment. Such a treatment will be given below.

Methods for the linear production of the configurations n_3 will then be attached to this. Schroeter² was the first to give such a procedure for a special class of configurations; it can also be applied to many other configurations. We shall see that for most configurations n_3 a linear construction is possible. The construction methods that we shall give cannot always be carried out, but they fail only in a few cases.

§1. Definition

A plane configuration n_3 (Cf. n_3) is a plane geometric figure consisting of n lines and n points so situated with respect to one another that every line contains three points and three lines pass through every point.

We denote the points of the configuration by n signs, for example by the numerals $1, 2, \dots, n$, and write these numerals in n columns, each of which is to represent a line of the configuration, by containing the three numerals that denote the points lying on the line. In this way we arrive at a schematic representation of the configuration. Each configuration can be denoted in many ways. Apart from the fact that it is immaterial how we arrange the columns of the scheme and the elements (numerals) inside any one column, the order in which we denote the points of the configuration by the numerals $1, 2, \dots, n$ is arbitrary. We shall therefore regard two configurations as the same if the scheme of one passes into that of the other by a substitution of the signs $1, 2, \dots, n$.

From the definition of configurations n_3 the following properties result for their notation:

1. Every column contains three elements.
2. Every element occurs in three columns.

¹Martinetti, "Sulle configurazione piane 4_3 ," Section 3, Annali di matematica pura ed applicata, Series II, vol. 15.

²Schroeter, "Über lineare Constructionen zur Herstellung der Configurationen n_3 " and "Über die Bildungsweise und geometrische Construction der Configurationen 10_3 ," Nachrichten der Königl. Gesellschaft der Wissenschaften zu Göttingen, 1888 and 1889.

3. A column may not contain one and the same element more than once.
4. Two columns may not agree in more than one element.

Every grouping of n elements into n columns that has these four properties will be called the scheme of a configuration n_3 or, when no misunderstanding is possible, simply a configuration n_3 ; the elements will also be called configuration-points and the columns configuration-lines, even if the scheme does not owe its origin to a geometric configuration. Conversely, starting from an arbitrary such scheme, we shall now investigate how geometric configurations corresponding to the given scheme can be produced.

§2. General decomposition into horizontal rows

A property that belongs not only to configurations n_3 but to all configurations n_μ helps us to a construction valid for all configurations n_3 ; we shall therefore prove the presence of this property in general for configurations n_μ .³

Theorem. In the scheme of every configuration n_μ , the elements inside the columns can be arranged in such a way that every horizontal row⁴ contains each of the elements $1, 2, \dots, n$ once, and hence only once.

To prove this theorem, we first show that one can associate the points and lines of the configuration in such a way that to each point there corresponds a line passing through it. Suppose that such an association has already been made with a number of points and lines. Let the complex of these lines be A , the complex of these points $\mathfrak{A}$, the complex of the remaining lines B , and the complex of the remaining points $\mathfrak{B}$. Then to every point of $\mathfrak{A}$ there corresponds a line of A passing through it, and our task plainly consists in enlarging the complex A so that it comprises all lines of the configuration. Then $\mathfrak{A}$ will contain all points of the configuration, and the complexes B and $\mathfrak{B}$ will disappear.

If a line b of B contains a point b of $\mathfrak{B}$, then we can associate b and b with one another. The point b thereby passes from $\mathfrak{B}$ into $\mathfrak{A}$, and the line b passes from B into A . This procedure is not applicable if all lines of B contain only points of $\mathfrak{A}$.

In that case we denote by a_0 all points of $\mathfrak{A}$ that lie on a line of B ; when distinction is needed we attach a second index. The corresponding lines, which must all belong to A , are denoted by α_0 . The points of $\mathfrak{A}$ lying on these lines but not occurring among the points a_0 are denoted by a_1 , their corresponding lines by α_1 ; the points of $\mathfrak{A}$ lying on these lines and not belonging to a_0 or a_1 are denoted by a_2 , their corresponding lines by α_2 , and so on. Since n is finite, we must finally arrive at points a_k whose corresponding lines α_k contain only points of $\mathfrak{A}$ already included among $a_0, a_1, \dots, a_k$.

We divide the complex $\mathfrak{A}$ into two complexes $\mathfrak{A}^0$ and $\mathfrak{A}'$, the first containing all points $a_0, \dots, a_k$, while the other consists of the remaining points of $\mathfrak{A}$. If the lines corresponding to the points of $\mathfrak{A}^0$ are assigned to a complex A^0 , and the lines corresponding to the points of $\mathfrak{A}'$ to a complex A' , then A^0 and A' together form the complex A .

Assume that a line of the complex A^0 , say α_i with $i \leq k$, contains a point of the complex $\mathfrak{B}$, say b' . If a'_i is the point corresponding to α_i , then it is contained in at least one of the lines α_{i-1} ; let α'_{i-1} be such a line, and let a'_{i-1} be the point corresponding to it. Then a'_{i-1} is

³By a configuration n_μ we here mean a grouping of n elements in n columns in which every column (line) contains μ elements (points), every element occurs in μ columns, and which also satisfies the third and fourth of the conditions set up for configurations n_3 .

⁴We imagine the columns of the configuration written side by side so that beside the k -th element of any column stands the k -th element of the next column.

contained in at least one of the lines α_{i-2} ; let α'_{i-2} be such a line. Continuing in this way we arrive at a sequence of points and lines

$$b', \alpha_i, a'_i, \alpha'_{i-1}, a'_{i-1}, \dots, \alpha'_0, a'_0, b'',$$

whose last member b'' is one of the lines of the complex B containing the point a'_0 . This sequence has the following properties. Every point, with the exception of b' , is associated with the preceding line and lies on it. But every point also lies on the following line. We can therefore change the association by making each point correspond to the following line. The points that previously belonged to $\mathfrak{A}$ remain in it; likewise all lines previously in A remain in A . But now the point b' , to which the line α_i is associated, passes from $\mathfrak{B}$ into $\mathfrak{A}$, and the line b'' , to which the point a'_0 belongs, passes from B into A .

It remains only to consider the case where no line of the complex A^0 contains a point of the complex $\mathfrak{B}$. The points of $\mathfrak{B}$, since they also lie on no line of B , can then occur only on the lines of the complex A' . But this too is impossible. The points of $\mathfrak{A}'$, as follows immediately from their definition, occur neither on the lines of B nor on those of A^0 ; they are therefore confined to the lines of A' . Since $\mathfrak{A}'$ contains just as many points as A' contains lines, while each point occurs μ times in the configuration and each line contains μ points, it follows that the lines of A' contain only the points of $\mathfrak{A}'$. Thus the complex B , and likewise $\mathfrak{B}$, is not present at all; the desired association of lines and points has already been reached.

We now place, in each of the columns denoting the lines of the configuration, that element in first position which denotes the point associated with the line. The first horizontal row of the configuration scheme will then contain every element once. The scheme that remains after removal of this row plainly denotes a configuration $n_{\mu-1}$, for which the theorem proved in general for configurations n_μ must again hold. We can therefore arrange the elements inside the columns of this scheme so that its first horizontal row, the second of the original scheme, contains every element once. It is immediate that the elements of the remaining horizontal rows, which always form a configuration, can be arranged so that each of those rows also contains every element once. The theorem stated above is thereby completely proved.

We apply the method just given to the following configuration 20₃:

Cf. 20_3:

1	1	1	2	2	2	3	3	4	4	5	5	6	6
3	4	8	9	11	15	5	15	7	12	7	10	7	8
10	9	17	16	14	18	13	19	13	16	14	18	20	10
6	8	9	11	11	12								
14	12	17	15	16	13								
19	19	20	20	17	18								

If we associate with the listed lines the points

$$1, 9, 2, 3, 19, 4, 16, 5, 10, 7, 8, 14, 17, 11, 12,$$

then these lines form the complex A and the corresponding points form $\mathfrak{A}$. The complex B consists of the lines

$$\begin{array}{cccccc} 1 & 1 & 2 & 3 & 11 \\ 3 & 4 & 11 & 12 & 16 \\ 10 & 9 & 14 & 19 & 17, \end{array}$$

and $\mathfrak{B}$ consists of the points 6, 13, 15, 18, 20.

All points lying on the lines of B belong to $\mathfrak{A}$; these are the points a_0 . Among the points a_1 one now has the points 5 and 7. The points 13, 18, 15, 6 belonging to $\mathfrak{B}$ lie respectively on the

lines

$$\begin{array}{cccc} 4 & 2 & 11 & 6 \\ 7 & 5 & 15 & 8 \\ 13 & 18 & 20 & 10. \end{array}$$

If we associate them with those lines, and instead associate with the points 4, 2, 11, 8 the lines

$$\begin{array}{cccc} 1 & 2 & 11 & 8 \\ 4 & 11 & 16 & 12 \\ 9 & 14 & 17 & 19, \end{array}$$

then B contains only the line $\begin{array}{c} 1 \\ 3 \\ 10 \end{array}$, and $\mathfrak{B}$ only the point 20. Further, one has

$$\begin{array}{lcl} a_0 & : & 1, 3, 10, \\ a_1 & : & 5, 8, 13, 17, 18, \\ a_2 & : & 2, 4, 7, 9, 12, 14, 15, 19, \\ a_3 & : & 6, 11, 16. \end{array}$$

The point 20 lies on the line corresponding to point 17; point 17 lies on the line corresponding to point 1; and point 1 lies on the line belonging to B . If we now associate with the points 20, 17, 1 the lines

$$\begin{array}{ccc} 9 & 1 & 1 \\ 17 & 8 & 3 \\ 20 & 17 & 10, \end{array}$$

then the desired association has been completely carried out.

Putting in each column, in first place, the element denoting the point associated with the corresponding line, and then deleting the first horizontal row of the scheme, one obtains three polygons

$$1, 9, 17, 16, 2, 15, 3, 10, 8; \quad 4, 7, 14, 11, 20, 6, 19, 12; \quad 5, 13, 18,$$

which compose the configuration.

Further decompositions of the configuration can easily be derived from this one. If we also arrange the second and third horizontal rows so that each contains all elements – and, if arrangements are regarded as the same when they can be carried into one another by interchanging columns and horizontal rows, this is possible in four ways – then, after deleting the second or third horizontal row, new representations of the configuration as a cycle of polygons result. The four essentially different possible arrangements are the following:

I.

$$\begin{array}{cccccccccccccccccccc} 4 & 20 & 11 & 9 & 18 & 19 & 1 & 6 & 17 & 13 & 5 & 2 & 15 & 7 & 14 & 8 & 16 & 3 & 12 & 10 \\ 1 & 9 & 17 & 16 & 2 & 15 & 3 & 10 & 8 & 4 & 7 & 14 & 11 & 20 & 6 & 19 & 12 & 5 & 13 & 18 \\ 9 & 17 & 16 & 2 & 15 & 3 & 10 & 8 & 1 & 7 & 14 & 11 & 20 & 6 & 19 & 12 & 4 & 13 & 18 & 5 \end{array}$$

II.

$$\begin{array}{cccccccccccccccccccc} 4 & 20 & 11 & 9 & 18 & 19 & 1 & 6 & 17 & 16 & 8 & 14 & 7 & 15 & 2 & 5 & 13 & 3 & 12 & 10 \\ 1 & 9 & 17 & 16 & 2 & 15 & 3 & 10 & 8 & 4 & 12 & 19 & 6 & 20 & 11 & 14 & 7 & 5 & 13 & 18 \\ 9 & 17 & 16 & 2 & 15 & 3 & 10 & 8 & 1 & 12 & 19 & 6 & 20 & 11 & 14 & 7 & 4 & 13 & 18 & 5 \end{array}$$

III.

$$\begin{array}{cccccccccccccccccccc} 4 & 20 & 11 & 9 & 18 & 19 & 1 & 6 & 17 & 16 & 8 & 14 & 7 & 15 & 2 & 5 & 13 & 10 & 12 & 3 \\ 1 & 9 & 17 & 16 & 2 & 15 & 3 & 10 & 8 & 4 & 12 & 19 & 6 & 20 & 11 & 14 & 7 & 5 & 18 & 13 \\ 9 & 17 & 16 & 2 & 15 & 3 & 10 & 8 & 1 & 12 & 19 & 6 & 20 & 11 & 14 & 7 & 4 & 18 & 13 & 5 \end{array}$$

IV.

4	20	11	9	18	19	1	6	17	13	5	2	15	7	14	8	16	10	12	3
1	9	17	16	2	15	3	10	8	4	7	14	11	20	6	19	12	5	18	13
9	17	16	2	15	3	10	8	1	7	14	11	20	6	19	12	4	18	13	5

Deleting in these schemes first the second and then the third horizontal row gives the following eight new decompositions of the configuration into polygons:

- 1) 1 10 5 14 19 3 13 7 6 8 12 18 15 20 17; 2 11 16 4 9.
- 2) 1 3 5 7 20 9 16 12 13 4; 2 14 6 10 18; 8 19 5 11 17.
- 3) 1 10 5 7 20 17; 2 14 6 8 19 3 13 4 9; 11 16 12 18 15.
- 4) 1 3 5 14 19 15 20 9 16 4; 2 11 17 8 12 13 7 6 10 18.
- 5) 1 10 18 15 11 16 12 13 4 9 2 14 6 8 19 3 5 7 20 17.
- 6) 1 3 13 7 6 10 5 14 19 15 20 9 16 4; 2 11 17 8 12 18.
- 7) 1 10 18 15 20 17; 2 11 16 4 9; 3 5 14 19; 6 12 13 7.
- 8) 1 3 13 4; 2 14 6 10 5 7 20 9 16 12 18; 8 19 15 11 17.

Further decompositions can be derived from these in the same way.

§3. Closed traversals

The proof just given is general and also includes the case where the configuration n_μ decomposes into several configurations. The latter case occurs, for example, whenever the complex A' disappears.

If ν horizontal rows are omitted from the configuration n_μ , then a configuration $n_{\mu-\nu}$ remains, which of course may also decompose. For $\nu = \mu - 2$ one obtains a polygon or a cycle of polygons, each side of which still contains $\mu - 2$ additional points of the configuration, while through each vertex there still pass $\mu - 2$ lines. We see from this that every configuration n_μ can be traversed in several closed circuits in such a way that every line occurs once as a side and every point once as a vertex.

If the given configuration n_μ decomposes, then every configuration n_ν contained in it, with $\nu < \mu$, must also decompose. But the converse of this theorem is not valid. A configuration n_μ may be completely connected, so that one can pass from any one of its points to any other by proceeding along configuration-lines, and nevertheless every configuration $n_{\mu-1}$ contained in it may decompose. Such cases already occur for configurations n_3 . Thus not every connected configuration n_3 can be traversed in a single closed circuit; rather, for any given number k one can form configurations n_3 that do not decompose but that cannot be traversed in fewer than k circuits.

§4. Arrangement of the complexes

In the following investigations, where we deal only with configurations n_3 , we assume that they do not decompose.

If the elements in the scheme of a configuration n_3 are arranged so that every horizontal row contains every element, then this property is preserved when the horizontal rows are interchanged; after such an interchange the scheme also denotes the same configuration as before. We now put an arbitrarily chosen horizontal row in first place; the two others denote a number of polygons.

To make this still clearer, and keeping the order of the elements within the columns unchanged, we arrange the columns as follows. We choose any column as the first, and determine the following columns in such a way that each second element of a column is identical

with the third of the preceding one. This procedure can be continued until we arrive at a column whose third element is identical with the second of the initially chosen column. The totality of columns, or lines, obtained in this way is called a complex of the configuration. If the complex does not yet comprise all columns of the configuration, we proceed to form a second complex by again taking an arbitrary one of the remaining columns and determining each following column in the same way as in the first complex. If the configuration is still not exhausted by this second complex, we form a third, fourth, and so on. That such a mode of formation is possible follows at once from the fact that every horizontal row of the scheme contains every element once.

If the first horizontal row of the configuration is omitted, one polygon remains from each complex. The second, and likewise the third, horizontal row of the complex at the same time gives the order in which the vertices of the polygon follow one another. This property is preserved when the columns of a complex are cyclically interchanged; no other interchange is permitted.

Let the number of complexes be k . We arrange them as follows. First we determine an arbitrary complex as the last one; call it C_k , its first horizontal row R_k , and the polygon represented by the two other rows P_k . If the configuration does not consist only of this complex, then not all elements of P_k can be contained in R_k ; at least one of them must occur in the first row of another complex. Otherwise the complex C_k would by itself already form a configuration.

We place such a complex in the next-to-last position and make a cyclic interchange of its columns so that the first element of the last column belongs to the elements contained in P_k . Call the complex thus determined C_{k-1} ; let R_{k-1} be its first horizontal row and P_{k-1} the polygon contained in it. If $k > 2$, not all points of P_{k-1} and P_k can be represented in the rows R_{k-1} and R_k ; at least one of them must occur in the first row of another complex. We choose such a complex as C_{k-2} and, by cyclically interchanging its columns, arrange that the element of P_{k-1} or P_k contained in R_{k-2} becomes the first element of the last column of C_{k-2} . It is clear how this procedure is to be continued, and that it leads to complexes $C_1, C_2, \dots, C_k$ with first horizontal rows $R_1, R_2, \dots, R_k$ and polygons $P_1, P_2, \dots, P_k$, having the property that the last point of R_i ($i < k$) occurs as a point of the second and third row only in one of the following polygons $P_{i+1}, \dots, P_k$.

§5. Normal notation

After the columns of the configuration have been arranged in this way, we replace the elements of the first horizontal row of the scheme, in order, by the signs $t_1, t_2, \dots, t_n$, and those of the second row by $p_1, p_2, \dots, p_n$. Each element of the third row is to be replaced by the sign that was introduced for the same element in the second row. We further denote by r_i ($i = 1, 2, \dots, n$) the column whose first element is t_i , and by p_i the column whose first element denotes the point also represented by the sign p_i . The notation introduced for elements and columns will also be used for the points and lines themselves. Thus each point and each line of the configuration receives two names.

If α_i ($i = 1, 2, \dots, k$) is the number of lines belonging to the complex C_i , we have the following scheme for the configuration:

$$\begin{array}{cccccc} \text{Cf. } n_3 & & & & & \\ t_1 & t_2 & \cdots & t_{\alpha_1} & t_{\alpha_1+1} & \cdots \\ p_1 & p_2 & \cdots & p_{\alpha_1} & p_{\alpha_1+1} & \cdots \\ p_2 & p_3 & \cdots & p_1 & p_{\alpha_1+2} & \cdots \end{array}$$

and analogously for the remaining complexes, up to the last one.

We now fix the following normal order for the points and lines of the configuration:

$$p_1, r_1, p_2, r_2, \dots, p_n, r_n.$$

This order is always being referred to when, of two elements (points or lines) of the configuration, one is said to precede or follow the other. Expressions such as “the first point,” “the last line of a polygon,” and the like are to be understood accordingly.

§6. Construction after $3n - 1$ conditions

The configuration n_3 can be regarded as a grouping of n lines $r_1, r_2, \dots, r_n$ and n points $p_1, p_2, \dots, p_n$ that arises by subjecting these initially arbitrary elements, one after another, to $3n$ conditions, not always independent, as if n lines and n points in the plane were moved until they assumed the position prescribed by the configuration. We first seek to produce a figure in which only $3n - 1$ of the $3n$ conditions are satisfied: for the time being we omit the requirement that the line r_n contain the point t_n . Notice that the omitted condition is not distinguished from the others. Since we placed an arbitrary complex last, and since any cyclic interchange was allowed among the lines of that complex, any line of the configuration can occupy the last place; hence r_n is an arbitrary line of the configuration. Since the horizontal rows of the scheme could also be interchanged arbitrarily, t_n is an arbitrary point of the line r_n .

We construct the figure in question by determining the points and lines in the normal order

$$p_1, r_1, p_2, r_2, \dots, p_n, r_n$$

and ensuring that the remaining $3n - 1$ conditions of the configuration are fulfilled. It is easy to see that this is possible. Suppose that the elements

$$p_1, r_1, p_2, r_2, \dots, p_i, r_i$$

have already been constructed. The point p_{i+1} is now to be determined.

There are two cases. If p_{i+1} is not the first point of one of the polygons $P_1, \dots, P_k$, then the configuration requires it to lie on the lines r_i, r_{i+1}, p_{i+1} . Of these lines r_{i+1} is not yet constructed, whereas r_i is; for p_{i+1} both cases are possible. Accordingly, depending on whether p_{i+1} belongs to the lines already constructed or not, the point p_{i+1} is to be placed at the intersection of the lines p_{i+1} and r_i , or it is to be taken as an arbitrary point of r_i .

If p_{i+1} is the first point of a polygon, and p_j the last point of the same polygon, then p_{i+1} must lie on the lines r_{i+1}, p_j, p_{i+1} , of which at most p_{i+1} has already been constructed. According to whether this is the case or not, p_{i+1} is to be taken arbitrarily on the line p_{i+1} , or altogether arbitrarily.

In constructing the following line r_{i+1} , two cases are again to be distinguished. If r_{i+1} is not the last line of a polygon, it must contain the points $p_{i+1}, p_{i+2}, r_{i+1}$. Of these, the first has already been found and the second has not. Depending on whether the point r_{i+1} belongs to the already constructed points or not, r_{i+1} must be the joining line of p_{i+1} and r_{i+1} , or an arbitrary line through the former point.

If r_{i+1} is the last line of one of the polygons $P_1, P_2, \dots, P_k$, and r_m is the first line of the same polygon, then r_{i+1} must contain the points p_n, r_{i+1}, p_{i+1} . The points p_n and p_{i+1} have already been determined, but r_{i+1} has not; for, in our arrangement of the configuration-lines, those points that lie on the last line of a polygon but do not occur there as vertices of that polygon must be vertices of one of the following polygons. Thus r_{i+1} must be constructed as the joining line of p_n and p_{i+1} .

Finally, if p_m is the first point of the last polygon and we construct r_n as the joining line $p_m p_n$, the constructed figure will satisfy $3n - 1$ conditions of the configuration.

§7. The last condition

In order to satisfy the last condition of the configuration as well, we seek, in the sequence of elements

$$p_1, r_1, p_2, r_2, \dots, p_n, r_n,$$

the last one that, after the preceding elements have been constructed and the last configuration-condition has been ignored, is not yet completely determined. We then try to dispose of this element in such a way that this condition also is fulfilled when the construction of the configuration is completed by the procedure described in §6. It is clear that the element sought belongs to the last polygon, as vertex or side, since the first point of this polygon, p_m , is not completely determined by the preceding elements. The element sought may be partially determined by what precedes it: if it is a point, it may be forced to lie on an already constructed line; if it is a line, it may be forced to pass through an already constructed point. It is also possible, however, that after the construction of the preceding elements it is still entirely undetermined.

This last case, which by the discussion of §6 can occur only when p_m , the first point of the last polygon, is the element sought, can be avoided. Since the point t_{m-1} must be a vertex of the last polygon, a cyclic interchange of the vertices and sides of the latter can be made so that $p_m = t_{m-1}$. Then p_m lies on the already constructed line t_{m-1} .

Let the element sought be a point p_i , with $i > m$. If, in the manner of §6, we construct the points and lines

$$p_1, r_1, p_2, r_2, \dots, p_{i-1}, r_{i-1},$$

and denote their totality by A , while the rest are collected under the symbol B , then the following is easily seen. The line p_i belongs to B ; for the point p_i lies on the lines r_{i-1} and p_i , of which the first belongs to A . Contrary to the hypothesis, it would already be completely determined if p_i also belonged to A . If now

$$p_i = t_k, \quad p_k = t_l,$$

and $k > i$, then the elements

$$t_k, p_{k+1}, r_{k+1}, \dots, p_n, r_n$$

with the exception of the point t_k all belong to A .

This is seen as follows. Suppose that a point t_j different from t_k , with $j > i$, belongs to B , and that $t_j = p_l$. Then $l > i$; $l = i$ is impossible, because $p_l = p_i$ would imply $t_j = t_k$. If $l > j$, then the line r_j , which contains the points t_j, p_j, p_{j+1} , would still not be completely determined after all preceding elements $p_1, r_1, \dots, p_j$ had been constructed. This is impossible, since p_i was supposed to be the last such element and r_j follows p_i . The only remaining possibility is $j > l > i$. Since r_j belongs to B and $r_j = p_l$, $j > l > i$, the sequence

$$t_i, p_{i+1}, r_{i+1}, p_{i+2}, \dots, p_n, r_n$$

contains before t_j an element different from t_k , namely p_l , that belongs to B .

Similarly, from the existence in the sequence $t_i, p_{i+1}, \dots, p_n, r_n$ of a line belonging to B one can infer the existence of a point, likewise belonging to B and different from t_k , which stands before the line in that sequence. Indeed, if $p_j = r_l$ is the line in question, then through p_j pass the lines r_{j-1}, p_j, r_j . Since $p_j = r_l$ is to occur in the sequence and belong to B , we first have $j > i$ and $l > i$. Since the point p_j is to be determined after the preceding elements have been constructed, because $j > i$, the line $p_j = r_l$ on which it lies must precede it in the normal order. Hence $j > l$. From $j > i$ it follows that p_j belongs to B ; from $j > l$ it follows that the point

$p_j = r_l$ stands before the line p_j in the sequence. Finally r_l cannot be identical with t_k , since $l = k$ would imply $j = i$, whereas $j > i$.

Thus every element that belongs to the sequence $t_i, p_{i+1}, \dots, p_n, r_n$, is different from t_k , and belongs to B , must be preceded in that sequence by an element with the same three properties. This gives a contradiction, since only finitely many elements are involved.

If the last condition of the configuration is now also taken into account, it follows from the foregoing that the elements

$$t_k, p_{k+1}, r_{k+1}, \dots, p_n, r_n, p_i, p_{i+1}, \dots$$

can be obtained in this order as joining lines and intersections of previously determined elements. For instance,

$$p_m r_n = r_n, (r_n, p_n) = p_n, \quad p_n r_{n-1} = r_{n-1}, \quad (r_{n-1}, p_{n-1}) = p_{n-1}, \dots$$

On the other hand, as soon as p_i has been assumed on the line r_{i-1} , one obtains $p_i r_i = r_i$, and analogously the following points and lines up to the element denoted by t_k .

The configuration still requires the points $t_k = p_i, p_k, p_{k+1}$, or respectively t_n, p_n, p_m if $k = n$, to lie on a line. Put first the joining line $p_{k+1} p_k = r'_k$ and the intersection $(r'_k, r_{i-1}) = p'_i$. As p_i traverses the line r_{i-1} , the point p'_i describes a projective point row on the same line. For all conditions of the configuration to be fulfilled, $p'_i = p_i$ must hold. Thus we arrive at the problem of finding the double elements of two incident point rows. It is well known that there are two such elements, and determining them usually requires a quadratic construction.

The cases in which p_i is identical with p_m , or in which the last element not completely determined by the preceding elements is a line, are dealt with in quite a similar way. A discussion of these cases may therefore be omitted.

Linear Constructions

§8. General problem

We have seen that the construction of any Cf. n_3 can be carried out by a number of linear operations and a single quadratic one. In special cases, however, the task of finding the double elements of two incident point rows, or of two pencils of rays, reduces to linear constructions; in that case we are able to construct the Cf. n_3 in the most general way by straightedge alone. Even when the procedure given in §§6–7 makes a quadratic construction unavoidable, it is not excluded that another procedure may require only linear operations.

If a general linear construction of the Cf. is impossible, it is nevertheless possible in most cases to obtain certain forms of the required Cf. by linear construction. Schroeter first gave such constructions for those Cff. which can be regarded as polygons in which the first, second, and fourth vertices always lie on one line. Schroeter's procedure can also be used for many other Cff.; he himself used it to obtain those Cff. 10_3 for which a general linear construction did not succeed. In what follows, without imposing any special hypothesis on the Cff. under consideration, a procedure will be given by which one arrives at their linear construction. For each Cf. to which this procedure is to be applied, a separate examination is nevertheless needed, since in certain cases the procedure becomes illusory.

§9. The two remaining free elements

In order to arrive at a general construction of the Cf. n_3 , we previously looked in the sequence of its elements

$$p_1, r_1, p_2, r_2, \dots, p_n, r_n$$

for an element that was not yet completely determined by the preceding elements, but that, together with them, determined all the following elements. The element sought is either a point p_i or a line r_i ; for the moment we shall not distinguish the two cases, and shall denote the element by e_i . The case previously avoided, in which e_i remains completely undetermined after the construction of the preceding elements, is favorable for the present purpose; but since it is easily disposed of after the other cases are considered, we exclude it for the moment.

We now look, in the sequence of elements preceding e_i , for an element which is not yet completely determined by the preceding elements, but after whose construction the following elements up to e_i , excluding e_i itself, are uniquely determined. It is clear that there must always be one and only one such element. We call it e_h , and express by the number h that it is identical with either p_h or r_h . After the preceding elements have been constructed, e_h may still be completely undetermined. We again exclude this case temporarily, since it too will be easy to settle later.

The element e_i always belongs to the last polygon, while e_h belongs either to the last or to the next-to-last polygon. If it belongs to the next-to-last, then e_i must be the first point of the last polygon. This property remains valid for the Cf. if we cyclically permute the lines of the last complex. We again make such a cyclic permutation so that p_m , the first point of the last polygon, becomes identical with t_{m-1} .

We now construct the elements preceding e_h by the method of §6, and ask whether and how it is then still possible to complete the Cf. using only the straightedge. We denote by A the totality of elements preceding e_h ,

$$p_1, r_1, p_2, r_2, \dots, p_{h-1}, r_{h-1} \quad (\text{or } p_h),$$

and by B the remaining elements

$$(p_h), r_h, p_{h+1}, r_{h+1}, \dots, p_n, r_n.$$

By our assumptions, e_h is incident with exactly one element of A ; the other two elements incident with e_h therefore belong to B . Of these two, one is always the element immediately following e_h ; the other will be denoted by η^h . Of the elements incident with e_i , exactly one must precede e_i ; of the other two, one immediately follows e_i , and the other will be denoted by η^i .

The elements e_h, e_i, η^h, η^i occur in the sequence

$$p_1, r_1, p_2, r_2, \dots, p_n, r_n$$

either in the order

$$e_h, e_i, \eta^h, \eta^i \quad (\text{Case I}),$$

or in the order

$$e_h, e_i, \eta^i, \eta^h \quad (\text{Case II}),$$

or in the order

$$e_h, \eta^h, e_i, \eta^i \quad (\text{Case III}).$$

It will be shown that in Cases I and II the Cf. can be completed linearly in arbitrarily many different ways, whereas in Case III no linear solution of the problem is possible in general.

The last line r_n contains the three points t_n, p_n, p_m . It may happen that r_n is identical with one of e_h, e_i , while p_m is identical with the other; then $r_n = \eta^h = \eta^i$. Although the construction of the Cf. in this case presents no difficulty, we shall always regard it as avoided. It must be shown that it can always be avoided.

More generally, whenever the present decomposition into complexes contains more than one complex, a mere change in the arrangement of the last complex can ensure that r_n is distinct from e_h and from e_i ; hence, since the point p_m identical with t_n cannot be completely determined after the preceding elements are constructed, it belongs to A . If the number k of complexes is greater than 1, at least one element of R_k , call it $v = p$, must be a vertex of one of the preceding polygons; otherwise C_k alone would form a Cf.. If e_h belongs to the last polygon, then $p = r$ belongs to A , and a cyclic permutation of the columns of C_k can make $p = r$ the first point of the last line. Thus that line belongs to A .

If e_h belongs to the next-to-last polygon, this cyclic permutation is not admissible, because the columns of the last complex were arranged so that $e_i = p_m$ should be identical with t_{m-1} . In this case r_n cannot be identical with e_i , since r_n contains $t_n, p_n, p_m = e_i$; it remains only to treat the case $r_n = e_h$. If this occurs, we interchange the second and third horizontal rows of C_k and reverse the order of the columns of this complex. We then obtain as P_k the same polygon as before, but in the reverse order. After a cyclic permutation of the columns of the last complex so that $e_i = p_m$ is identical with t_{m-1} , the element now denoted by e_i is the same as before, while the point now denoted by t_n is different from the previous one. Hence r_n and e_h are distinct, and t_n belongs to A .

§10. General auxiliary facts

For all cases the following observations apply.

1. The elements e_h, e_i, η^h, η^i are always distinct. Of e_h and η^h , one is a point and the other a line; the same holds for e_i and η^i .

2. Every element of B distinct from e_h, e_i, η^h, η^i is incident with the immediately preceding element, with the immediately following element unless it is r_n , and with an element of A .

Indeed, if e is such an element, it is incident with the immediately preceding element, since every element of the Cf. which is not the first point of a polygon has this property. Among the elements of B , only e_h and e_i can fail to have it. Since e is completely determined by preceding elements, it must be incident with two of them. Let e' be the first element, in the normal order, which is incident with e . Between e' and e there is another element incident with e , namely the element immediately preceding e . The element e' cannot be identical with e_h or e_i , and it cannot be identical with η^h or η^i either. If e' belonged to B , then e' would be incident with its immediate predecessor and, since it is not r_n , also with its immediate successor; together with the incidence with e , all three elements incident with e' would belong to B . Hence, if among the elements of B different from e_h, e_i, η^h, η^i there were an element not incident with any element of A , it would be preceded by another element with the same property. This is impossible, since B contains only finitely many elements. Thus every such e is incident with an element e' of A .

3. The latest among the four elements e_h, e_i, η^h, η^i in the sequence may be r_n . Then one of the points t_n, p_m is identical with one of e_h, e_i , and only one of these points can be so. If $p_m = e_h$ or e_i , then r_n belongs to A ; if $t_n = e_h$ or e_i , then e_h and e_i lie in the last polygon and are distinct from p_m , so that p_m belongs to A .

If η^h and η^i are distinct from r_n , say one of them is p_s or r_s , then the conditions of the Cf., including the last condition expressing the incidence of t_n with r_n , prescribe the construction

of the elements following η :

$$(r_s), p_{s+1}, r_{s+1}, \dots, p_n, r_n.$$

The elements $(\eta), p_{s+1}, r_{s+1}, \dots, p_n, r_n$ and also p_m must all belong to A . Hence the successive intersections and joining lines are obtained as before:

$$p_m r_n = r_n, \quad (r_n, p_n) = p_n, \quad p_n r_{n-1} = r_{n-1}, \quad \dots, \quad (r_{s+1}, p_{s+1}) = p_{s+1}.$$

Consequently, after all elements belonging to A have been constructed, all elements following η are determined if the last condition of the Cf. is also taken into account. We construct these elements, denote all elements constructed so far by A' , and the remaining ones by B' . In choosing e_h and e_i we must therefore ensure that the element η determined by them is incident with the immediately following element of A' ; if η^h or η^i is r_n , this following element is replaced by whichever of t_n, p_m belongs to A .

§11. Reduction of the cases

Assume first that Case I or II occurs. Once all elements belonging to A' have been constructed, the remaining elements can, after a completely arbitrary choice of e_i , still be determined so that only one of all the conditions of the Cf. remains unfulfilled. In these two cases none of the elements η^h, η^i lies between e_h and e_i . Every element between e_h and e_i is therefore incident, by §10,2, with its immediate predecessor, its immediate successor, and an element of A . Thus, once e_i is chosen, the elements between e_h and e_i , including e_h itself, are obtained in reverse order, each as an intersection or joining line of the following, already constructed element with an element of A .

By the definition of e_i , all elements following e_i are now determined and can be obtained by the procedure of §6. Those following η have already been constructed; if the remaining ones up to η are constructed, then $3n - 1$ of the $3n$ conditions of the Cf. are satisfied. The remaining condition is that, for an arbitrary choice of e_i , the element η will not in general be incident with the following element. In order for this condition also to hold, e_i must be a point or a tangent of a certain curve. The task is therefore to determine, by linear constructions, arbitrarily many points or tangents of this curve.

Case III is essentially different, because there η^h lies between e_h and e_i and η^h is incident with no element of A' .

In addition to the order in which e_h, e_i, η^h, η^i occur, we must distinguish their nature. Either

- a) $e_h = p_h, \quad e_i = p_i, \quad \eta^h$ and η^i are lines,
- b) $e_h = r_h, \quad e_i = r_i, \quad \eta^h$ and η^i are points,
- c) $e_h = p_h, \quad e_i = r_i, \quad \eta^h$ is a line, η^i a point,
- d) $e_h = r_h, \quad e_i = p_i, \quad \eta^h$ is a point, η^i a line.

Thus there are twelve cases, denoted

Ia	Ib	Ic	Id
IIa	IIb	IIc	IIId
IIIa	IIIb	IIIc	IIId.

§12. Case I

In Cases I and II, e_h cannot be the first point of the next-to-last polygon; otherwise $\eta^h = r_{m-1}$, $\eta^i = r_m$, and $e_i = p_m$, so that η^h would occur between e_h and e_i , which is Case III. In Case I, e_h cannot be the first point of the last polygon either, for then $\eta^h = r_n = \eta^i$, giving Case II.

We first treat Case Ia. Then $e_h = p_h$, $e_i = p_i$, with the corresponding incident elements denoted as above. Suppose the relevant indices satisfy $h < i < j < l$. The elements belonging to A' are already constructed. As shown in §11, after an arbitrary choice of p_i the remaining elements of B' may be constructed so that only the condition requiring the incidence of $p_l = r_1$ and p_{l+1} is not satisfied. One may instead impose that condition and leave another one out.

Following the prescriptions of the Cf., one obtains chains of intersections and joining lines from the chosen point p_i ; for arbitrary p_i the only condition left open is that the points

$$p_h = t_j, \quad p_j, \quad p_{j+1}$$

should lie on one line, namely the Cf.-line r_j .

If the point p_i moves in the plane, then the other elements of B' also move. The lines p_i, r_{i-1}, r_i rotate about fixed points, while the points $p_h = t_j, p_j, p_{j+1}$ run through fixed lines. Each of the corresponding pencils is projectively related to the corresponding point row. Thus the construction is reduced to the following problem.

Given are three points, the centers of three pencils of rays, and three lines, the carriers of three point rows. To each pencil a point row is projectively assigned. Through any point p of the plane there passes one ray of each pencil; to these three rays correspond three points in the assigned point rows. One must find those points p for which the three points thereby obtained lie on one line.

Every solution of this problem also solves the dual problem: find lines in the plane such that the three rays corresponding to their intersections with three given lines meet in a point. This dual problem gives Case Ib. Cases Ic and Id are obtained by the same reasoning. In Case Ic, the moving element is a line r_i rather than a point p_i ; the relevant points move on projectively related rows, and one is led to this form: given three lines carrying point rows and three further lines carrying point rows projectively assigned to the first three, find lines which cut the first three in points whose assigned points again lie on one line. Case Id gives the dual problem.

§13. Case II

We next treat Case IIa. Here $e_h = p_h$, $e_i = p_i$, while the two elements η^h, η^i occur in the reverse order. With the notation chosen in the text, all relevant elements preceding and following the two free choices belong to A' except the elements to be constructed from p_i . If p_i is chosen arbitrarily in the plane, then it determines all the missing elements by the successive intersection and joining operations prescribed by the Cf.. The only condition neglected is that the point $p_i = r_1$ should lie on the line r_1 .

As p_i varies, two of the determined lines describe pencils and two of the determined points describe point rows projectively related to those pencils. We arrive at the following problem. Given are two points, centers of two pencils, and two lines, carriers of two point rows; each row is projectively related to one of the pencils. These relations assign to each point of the plane a line, and conversely to each such line the same point. Through the point pass two rays, and to those rays correspond two points; their joining line is the line assigned to the given point. One must find those points which lie on their assigned lines. Equivalently, one may ask for those lines which pass through their assigned points. This gives the dual Case IIb.

Case IId differs only superficially from IIa and leads to the same problem; Case IIc is dual to IId.

§14. Case III

In Case IIIa the order is e_h, η^h, e_i, η^i . The construction can be split into three independent projective motions: one point runs through a point row, another point runs through a second point row, and a line rotates in a pencil. Taking into account the two still unfulfilled Cf.-conditions, one obtains the following problem.

Given are two projective point rows, one point row projective to a pencil, and two projective pencils. It is required to determine corresponding pairs of elements such that three points lie on one line and three lines pass through one point. Case IIIb leads to the same problem.

Similarly Case IIIc leads to this problem. Given are three pairs of lines $a, a_1; b, b_1; c, c_1$. Points of a are projectively assigned to points of a_1 , points of b to points of b_1 , and points of c to points of c_1 . One has to find corresponding points $a, a_1; b, b_1; c, c_1$ such that a, a_1, b are collinear and b_1, c, c_1 are collinear. Case IIId gives the dual problem.

§15. The excluded cases

It remains to discuss the two cases excluded in §9. If e_i is not determined at all by the preceding elements, then necessarily $e_i = p_m$. The point t_{m-1} must be a vertex of the last polygon. By cyclically permuting the lines of the last complex so that this vertex becomes the first vertex of the last polygon, and by changing the notation of the vertices and sides of that polygon accordingly, the previous point p_m obtains a later name $p_s = r_t$ with $s, t > m$. One sees that the situation is reduced to one of Cases IIa or IIc.

If e_h is not determined at all by the preceding elements, then it is the first point of the next-to-last or last polygon. If it is the first point of the next-to-last polygon, then a reordering of the complexes placing that complex last reduces the case to the previous one. If the line involved belongs to the last complex, then after one arbitrary joining line and a chain of the standard intersections and joins, the only remaining condition is again that three corresponding points should be collinear; this is the same problem as in Case Ia.

Finally suppose e_h is the first point of the last polygon and remains completely undetermined. We call this Case IV. Every element of the last polygon is incident with its two neighbors. In the present case e_h is also incident with a non-neighbor, denoted η_h , and e_i with a non-neighbor η_i . Among all vertices and sides of the last polygon, only e_h, e_i, η_h, η_i have the property of also being incident with a non-neighbor. This distinguished set is unchanged by cyclic permutation of the lines of the last complex.

Among these four elements are two points and two lines. Denote the points by e_1, e_2 , and the non-neighboring lines through them by η_1, η_2 . This notation refers to fixed elements of the Cf., independently of the position they acquire in a particular cyclic arrangement. By contrast, e_h, e_i, η_h, η_i denote, for each arrangement, the elements that play the corresponding roles. The order of identification is determined by the order in which e_1, e_2, η_1, η_2 occur.

If one of e_1, e_2 is made the first vertex of the last polygon, Case IV occurs; otherwise one obtains Case I, II, or III according as the first two distinguished elements have different or equal indices. If e_1 and η_1 are neighboring, then e_2 and η_2 cannot also be neighboring, and an appropriate cyclic permutation gives Case I or II. The same holds symmetrically. If neither e_1 lies on η_1 nor e_2 on η_2 , then no two of the four elements are neighbors, and cyclic permutations allow an ordering that avoids Case IV and, unless two trial arrangements both yield Case III, directly gives Case I or II. In the exceptional ordering, making one of the points the second point of P_k gives Case II. Thus Case IV can always be reduced to Case I or II.

§16. The cubic curve

In Cases I and II the desired element e_i must be a point or tangent of a curve of order or class three determined by the previous construction.

For Case Ia, let the centers of the three pencils be $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}$, and let the carriers of the three point rows be A, B, C , so that the pencil $\mathfrak{A}$ is projectively related to row A , and similarly for the other two. If p is a variable point of the plane and

$$p\mathfrak{A} = a, \quad p\mathfrak{B} = b, \quad p\mathfrak{C} = c,$$

and if a, b, c also denote the corresponding points on A, B, C , then we must find the points p for which a, b, c are collinear.

If p moves on a straight point row, then the rays a, b, c describe pencils and the points a, b, c describe point rows projectively related to each other. Thus one seeks triples of corresponding points on A, B, C that lie on one line. As p moves along the straight row, the line ab envelops a conic K , and the line ac envelops a conic K' . If a, b, c are collinear, their common line is a common tangent of the two conics. Apart from a , the two conics have at most three common tangents in general. Hence any straight line traced by p contains at most three points with the desired property; the locus is therefore a cubic curve.

If two points p_1, p_2 of this curve are known, then the line p_1p_2 contains a third point p_3 of the curve, which can be found linearly. This is reduced to constructing the fourth common tangent of two conics K, K' when three common tangents are known. Two of the known tangents are cut by the remaining tangents of each conic in projective point rows. Thus the points of one tangent are projectively related to those of the other in two ways. One double point is the intersection of the two known tangents; the other is obtained by a linear construction. The corresponding line is the fourth common tangent, and it determines the new point p_3 .

The corresponding lines $g = abc$ envelop a curve of class three. Just as two points of the cubic determine a third by a linear construction, two tangents of the class curve determine the third tangent through their intersection. The analogous constructions apply to Cases Ib, Ic, and Id. In Case II one obtains the same phenomenon from two pencils and two point rows: a moving point has an assigned line, and the desired points are precisely those that lie on their assigned lines. Once two such points on a line are known, the third can be found linearly.

The constructions presuppose that some elements of the curve are already known. This is so. In Case Ia the three centers $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}$ lie on the required curve. In Case II, $\mathfrak{A}$ and $\mathfrak{B}$ lie on it, and a third point is obtained from the intersection of A and B ; the corresponding line gives a tangent of the class curve. The third points on the sides of the triangle $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}$ are

$$(\mathfrak{B}C, A), \quad (C\mathfrak{A}, B), \quad (\mathfrak{A}\mathfrak{B}, C).$$

§17. Impossibility in Case III

The situation is quite different in the cases III. Considering the problem obtained in Case IIIc, one sees that construction of the required points by straightedge alone is impossible in general. This already follows from the fact that it can happen, and not merely in a special case, that no real system of such points exists at all.

Using the notation of §14, let $a, a_1; b, b_1; c, c_1$ be variable pairs of corresponding points on the six given lines. The lines aa_1 envelop a conic K , and the lines cc_1 envelop a conic K_1 . If the line b meets K and the line b_1 meets K_1 in real points, then each of these two lines is divided into two parts so that from one part real tangents to the associated conic are drawn, and from the other only imaginary tangents. If the projective correspondence between b and b_1

sends the segment of b belonging to real tangents entirely into the segment of b_1 belonging to imaginary tangents, then there is no pair of corresponding points b, b_1 from which real tangents go to both conics. This proves the assertion. The impossibility of a linear construction in Cases IIIa, IIIb, and IIIc is recognized in the same way.

§18. Conclusion

We have seen that every non-decomposing Cf. n_3 can be constructed by linear constructions with the addition of at most one quadratic construction. The construction given is completely general: every figure obtained by it is a Cf. of the required kind, and every Cf. of the required kind occurs among the figures obtainable by it.

For the second assertion, the construction must be understood with the special cases included. When two lines whose intersection is to be found coincide, an arbitrary point of that line is to be taken in place of the intersection; and when two points whose joining line is to be found coincide, an arbitrary line through them is to be taken.

Special cases must not be excluded. The problem of giving a general construction that produces all Cff. of the required kind in which distinct Cf.-points or Cf.-lines do not coincide is quite different from the one treated here. In many Cff. certain lines or points must in fact always coincide. One example is Schroeter's configuration

Cf. 10₃:

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1 1 1 2 2 3 3 4 5 8
2 4 6 4 5 6 7 6 7 9
3 5 7 8 9 8 9 10 10 10.
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If a Cf. n_3 is to be constructed linearly, then in decomposing it into polygons and arranging those polygons one must avoid Case III, since that case generally prevents the construction from being completed by straightedge alone. Because the elements of the schema can be arranged in many ways so that each horizontal row contains every element once, because either pair of horizontal rows may be used to form the polygons, and because the order of the polygons and of the lines inside them can be chosen in several ways, it will usually be possible to produce one of Cases I and II. Whether this is possible for every Cf., or whether there are Cff. for which every arrangement leads to Case III, could not be decided.

In most Cff. it seems possible, by different arrangements of the elements, to produce any desired one of Cases I, II, III. In the Cf. 20₃ treated in §2, the three arrangements displayed in the German source arise from one and the same decomposition into the polygons

1, 9, 17, 16, 2, 15, 3, 10, 8; 4, 7, 14, 11, 20, 6, 19, 12; 5, 13, 18.

Cases I and II led to the problem of finding, from certain known points of a uniquely determined cubic curve, further points by linear construction, and similarly for the dual problem. In general it suffices to know one point of the curve in order to derive arbitrarily many other points from it, covering the whole curve or one of its branches so that the distances between successive points are smaller than any prescribed length. In special cases, however, even if three points of a uniquely determined nonsingular cubic are known, it is impossible by linear construction to determine any other point of the curve. Thus for each Cf. the applicability of the methods given for linear construction still requires a special examination.

Vita

I was born, Ernst Steinitz, in the village called Laurahütte on 13 June 1871, to my father Sigismund, whose death I deeply mourn, and to my mother Augusta, née Cohn. After attending

the Friedrich Gymnasium in Breslau for ten years, I obtained the certificate of maturity in 1890. Soon thereafter I was admitted to the philosophical faculty and devoted eight semesters chiefly to mathematical studies, first in Breslau, then in Berlin, and then again in Breslau. I heard the distinguished teachers Baeumker, Dieterici, Hintze, Lipps, London, Meyer, Milch, Rosanes, Schmarsow, and Schroeter in Breslau; Aegidi, Delbrueck, Ebbinghaus, Frobenius, Fuchs, Grimm, Knoblauch, Kronecker, Planck, and Schwarz in Berlin. To all these men I offer my greatest thanks, above all to Jakob Rosanes, who, I gratefully acknowledge, supported my studies with the greatest goodwill.

Theses

1. The elements of number theory should already be taught at school.
2. The reintroduction of Latin as a written language would have many advantages for the mathematical sciences.
3. The geometry of n dimensions has no less reality as a mathematical discipline than the geometry of three dimensions.